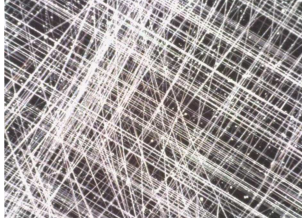
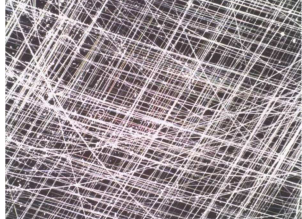
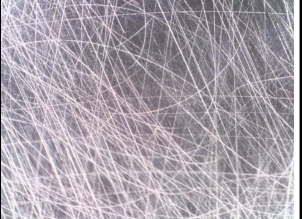
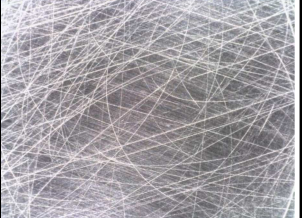
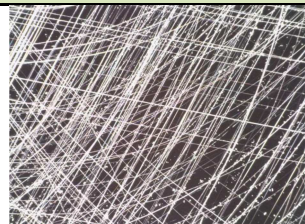
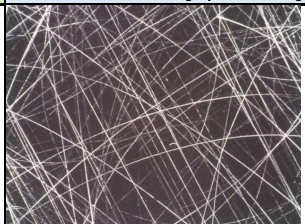
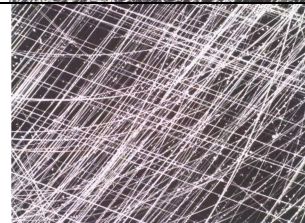
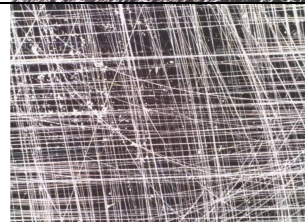
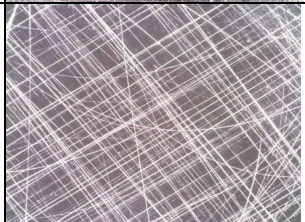
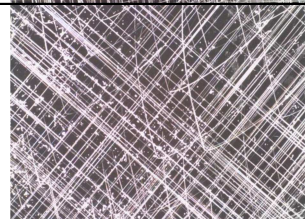
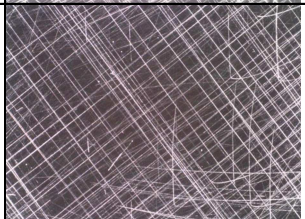


8.4 PMMA Polishing Test Results

Material: artificially abraded black PMMA panel/ hand polishing: 2 minutes/ machine polishing: 2 minutes/ microscopy magnification: 70 x

Test date : 16-17. Oct. 2024

No.	Type of polishing pastes	Before (abraded)	After		Remarks
			Polishing by hand , using suede microfiber cloth	Polishing by mini electric grinder	
1	Novus Polish #2 Fine Scratch Remover 		 Overall scratch reduction: good ✓ White haze(micro scratches): reduced	 Overall scratch reduction: good White haze(micro scratches): increased	Hand polishing surface visual under the light (unaided eye level) 
2	Novus Polish #3 (Heavy) Scratch Remover 		 Overall scratch reduction: bad White haze(micro scratches): increased	 Overall scratch reduction: bad White haze(micro scratches): increased	
3	Burnus Acrylic Glass Polish & Repair Paste 		 Overall scratch reduction: good ✓ White haze(micro scratches): reduced	 Overall scratch reduction: bad White haze(micro scratches): increased	Hand polishing surface visual under the light (unaided eye level) 
4	3M Perfect-it III Ultra Fine SE 		 Overall scratch reduction: bad White haze(micro scratches): reduced	 Overall scratch reduction: bad White haze(micro scratches): increased	
5	3M Perfect-it III 80349 Extra Fine Compound Polishing Paste 		 Overall scratch reduction: good ✓ White haze(micro scratches): reduced	 Overall scratch reduction: middle White haze(micro scratches): increased	Hand polishing surface visual under the light (unaided eye level) 

No.	Type of polishing pastes	Before (abraded)	After				Remarks
			Polishing by hand , using suede microfiber cloth		Polishing by mini electric grinder		
6	3M 09639 Finesse-it Sanding Paste 			Overall scratch reduction: good ✓ White haze(micro scratches): reduced		Overall scratch reduction: very good ✓✓ White haze(micro scratches): increased	Machine polishing surface visual under the light (unaided eye level) 
7	3M Perfect-it III Fast Cut Plus Extreme 			Overall scratch reduction: middle White haze(micro scratches): increased		Overall scratch reduction: good White haze(micro scratches): increased	The paste has sharp grains that the user can feel.
8	PolyWatch 			Overall scratch reduction: middle White haze(micro scratches): increased		Overall scratch reduction: very good White haze(micro scratches): increased	The paste texture is similar to liquid.
9	Commandant Cm4t Reiniger M4 			Overall scratch reduction: good ✓ White haze(micro scratches): reduced		Overall scratch reduction: good White haze(micro scratches): increased	

* Scales - Overall scratch reduction: bad/ middle/ good / very good
- White haze(micro scratches): increased/ reduced

* Light: white LED lamp

Notes

- 1) In the microscopy images: the black color is the original surface color of black PMMA/ abrasions and scratches are in the white color.
- 2) Hand polishing with pastes: tends to reduce shallow abrasions, while thicker abrasions remain. It doesn't make micro scratches on the background.
- 3) Machine polishing with pastes: tends to reduce shallow and thicker abrasions, working better than hand polishing. However, most of the time it makes micro-scratches on the background, causing a white haze.
- 4) Green marked: when scratch reduction is good, as well as white hazes are reduced.